

OFFICIAL LABORATORY TEST & COMPLIANCE CERTIFICATE

QUALANOD ARCHITECTURAL ANODIZING QUALITY CERTIFICATE

Class 20 & Class 25 Anodic Oxidation Layers for Marine Architectural Applications

Governing Standard:	QUALANOD European Quality Standard • ISO 7599 • ISO 2143	Issuance Date:	January 15, 2026
Accredited Testing Facility:	EuroVent Architectural Facade Labs (Notified Body)	Audit Cycle:	2026 – 2029 (Triennial Review)
Manufacturer & Facility:	AinZara Aluminum Factory, Industrial Area, Tripoli	Specimen Origin:	Alloy 6063-T6 / PA66 Polyamide

1. SCOPE OF CERTIFICATION & TEST METHODOLOGY

Comprehensive auditing and laboratory bath verification for sulfuric acid anodized aluminum extrusions. Assessment includes anodic oxide layer thickness, sealing quality, surface micro-hardness, and chemical resistance to alkaline and acidic exposure.

2. LABORATORY TEST RESULTS & CLASSIFICATION METRICS

Test Parameter / Clause	System / Substrate	Standard Target	Measured Value	Evaluation
Anodic Layer Thickness (Class 20)	Standard Exterior Class	Minimum $\geq 20.0 \mu\text{m}$	Measured $22.1 \mu\text{m}$ Average	PASS (Class 20)
Anodic Layer Thickness (Class 25)	Heavy Marine Grade	Minimum $\geq 25.0 \mu\text{m}$	Measured $26.8 \mu\text{m}$ Average	PASS (Class 25)
Sealing Quality: Admittance Test	ISO 2931 (1 kHz)	$< 20 \mu\text{S}$ / Thickness	Measured $11.2 \mu\text{S}$	PASS (Optimal Seal)
Sealing Quality: Dye Spot Test	ISO 2143 (Color Stain)	Grade 0 to 1 Acceptable	Grade 0 (Zero Absorption)	PASS (Impervious)
Nitric Acid Mass Loss Test	ISO 3210 Chemical Attack	$< 30 \text{ mg/dm}^2$ Mass Loss	Measured 14.5 mg/dm^2	PASS (High Resistance)
Surface Micro-Hardness	ISO 10074 Vickers Test	$> 350 \text{ HV } 0.05$	Measured 382 HV	PASS (Scratch Resistant)

3. REGULATORY COMPLIANCE & USAGE ATTESTATION

The anodizing line produces uniform, densely sealed anodic coatings with zero microporosity. Provides maximum protection against architectural abrasion, salt-laden sea mist, and intense solar ultraviolet radiation while preserving the natural metallic brilliance of the architectural alloy.

**ACCREDITED
DIRECTOR**

Dr. Eng. Marcus Von Berg, Ph.D.
Lead Auditor, Facade Testing Institute
Munich / EuroVent Network

LABORATORY

[SEAL]

**OFFICIAL SEAL OF CONFORMANCE
AIN ZARA TESTING BUREAU**

**DIRECTOR OF QUALITY
ASSURANCE**

Eng. Tarek Ben Mahmoud
Chief Technical Officer
Ain Zara Aluminum & Glass Industries

